



Chapter 5

Methods of Economic Analysis

Er. Shrutee Karna
Department of Civil Engineering

CONTENT

5 Methods of Economic Analysis

(12 hours)

5.1 Capital budgeting

5.2 Minimum attractive rate of return (MARR)

5.3 Economic analysis of single and multiple projects

5.3.1 Payback period (Simple and discounted)

5.3.2 Equivalent worth (Net present, annual, future and capitalized)

5.3.3 Rate of return (Internal and external)

5.3.4 Public Sector economic analysis (Benefit cost analysis)

5.3.5 Financial and economic analysis

5.4 Weighted average cost of capital (WACC)

5.5 Repeatability assumption and Co-Terminated assumptions

5.6 Multiple investment project alternatives (Dependent, independent and contingent)

Capital budgeting

- **Capital budgeting** is the formal process of using quantitative analysis techniques to evaluate, prioritize, and select long-term engineering projects that are technically viable and economically sound. This process is crucial because engineering decisions often involve substantial, irreversible capital expenditures that have long-term impacts on a company's profitability and strategic direction.
- Capital budgeting is the process of analyzing investment opportunities in long-term assets using economic evaluation techniques to determine whether the expected returns justify the initial cost.

Importance of Capital Budgeting in Engineering Economics

1. Optimal Allocation of Resources

Ensures that limited financial resources are invested in the most profitable and technically feasible projects.

2. Supports Long-Term Planning

Helps organizations plan future expansion, modernization, and infrastructure development systematically.

Capital budgeting

3. **Incorporates Time Value of Money**
Evaluates projects by considering that money available today is worth more than the same amount in the future.
4. **Reduces Investment Risk**
Systematic economic analysis minimizes the chances of selecting unprofitable or technically unsound projects.
5. **Profitability and Financial Stability**
Guides firms toward projects that generate higher returns and improve overall financial performance.
6. **Facilitates Objective Decision Making**
Provides quantitative techniques (NPW, IRR, AW) that allow engineers and managers to compare alternatives scientifically rather than relying on judgment alone.

Capital budgeting

Methods of Capital Budgeting in Engineering Economics

- a) Present Worth Method (PW / Net Present Value – NPV)
- b) Annual Worth Method (AW / Equivalent Annual Worth – EAW)
- c) Future Worth Method (FW)
- d) Internal Rate of Return (IRR)
- e) External Rate of Return (ERR) / Modified Internal Rate of Return (MIRR)
- f) Benefit–Cost Ratio (BCR) / Profitability Index (PI)
- g) Payback Period Method

Minimum attractive rate of return (MARR)

Minimum attractive rate of return (MARR) is the minimum acceptable rate of return established by an organization for evaluating and selecting investment projects. It represents the interest rate that must be exceeded for a project to be considered economically worthwhile.

Basis for Determining MARR

- i. Cost of borrowed capital
- ii. Expected return from alternative investments (opportunity cost)
- iii. Project risk
- iv. Inflation rate
- v. Organizational financial policy

Minimum attractive rate of return (MARR)

Role in Decision-Making

In Engineering Economics, the MARR is the "i" (interest rate) used in various evaluative methods:

- Net Present Value (NPV), Annual Value, Future Value: A project is acceptable if its NPV, or AV or $FV \geq 0$.
- Internal Rate of Return (IRR): A project is considered viable if its $IRR > MARR$.
- Ranking: When comparing multiple projects, MARR provides a consistent standard to prioritize the most profitable ones

Equivalent worth (Net present, annual, future and capitalized)

I. Present Worth (PW) Method

The Present Worth Method evaluates a project by converting all future cash inflows and outflows to their equivalent value at the present time (time zero) using a given interest rate.

In simple words, it answers:

“What is the value today of all future benefits and costs?”

In this method of comparison, the cash flows of each alternative will be reduced to time zero by assuming an interest rate i . Then, depending on the type of decision, the best alternative will be selected by comparing the present worth amounts of the alternatives. The sign of various amounts at different points in time in a cash flow diagram is to be decided based on the type of the decision problem.

Present Worth Method (PW)

In this method of comparison, the cash flows of each alternative will be reduced to time zero by assuming an interest rate i . Then, depending on the type of decision, the best alternative will be selected by comparing the present worth amounts of the alternatives. The sign of various amounts at different points in time in a cash flow diagram is to be decided based on the type of the decision problem. In a cost dominated cash flow diagram, the costs (outflows) will be assigned with positive sign and the profit, revenue, salvage value (all inflows), etc. will be assigned with negative sign. In a revenue/profit-dominated cash flow diagram, the profit, revenue, salvage value (all inflows to an organization) will be assigned with positive sign.

The costs (outflows) will be assigned with negative sign. In case the decision is to select the alternative with the minimum cost, then the alternative with the least present worth amount will be selected. On the other hand, if the decision is to select the alternative with the maximum profit, then the alternative with the maximum present worth will be selected.

Present Worth Method (PW)

REVENUE-DOMINATED CASH FLOW DIAGRAM :

A generalized revenue-dominated cash flow diagram to demonstrate the present worth method of comparison is presented

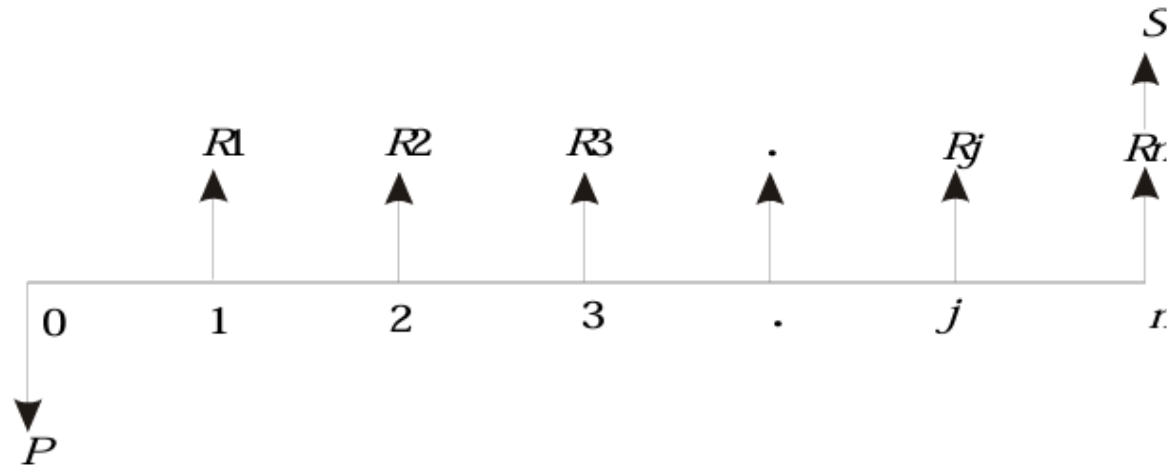


Fig. Revenue-dominated cash flow diagram.

Present Worth Method (PW)

P represents an initial investment and R_j the net revenue at the end of the j th year. The interest rate is i , compounded annually. S is the salvage value at the end of the n th year. To find the present worth of the above cash flow diagram for a given interest rate, the formula is

$$PW(i) = -P + R_1[1/(1+i)^1] + R_2[1/(1+i)^2] + \dots \\ + R_j[1/(1+i)^j] + R_n[1/(1+i)^n] + S[1/(1+i)^n]$$

In this formula, expenditure is assigned a negative sign and revenues are assigned a positive sign. If we have some more alternatives which are to be compared with this alternative, then the corresponding present worth amounts are to be computed and compared. Finally, the alternative with the maximum present worth amount should be selected as the best alternative

Present Worth Method (PW)

COST-DOMINATED CASH FLOW DIAGRAM:

A generalized cost-dominated cash flow diagram to demonstrate the present worth method of comparison is presented in Fig.

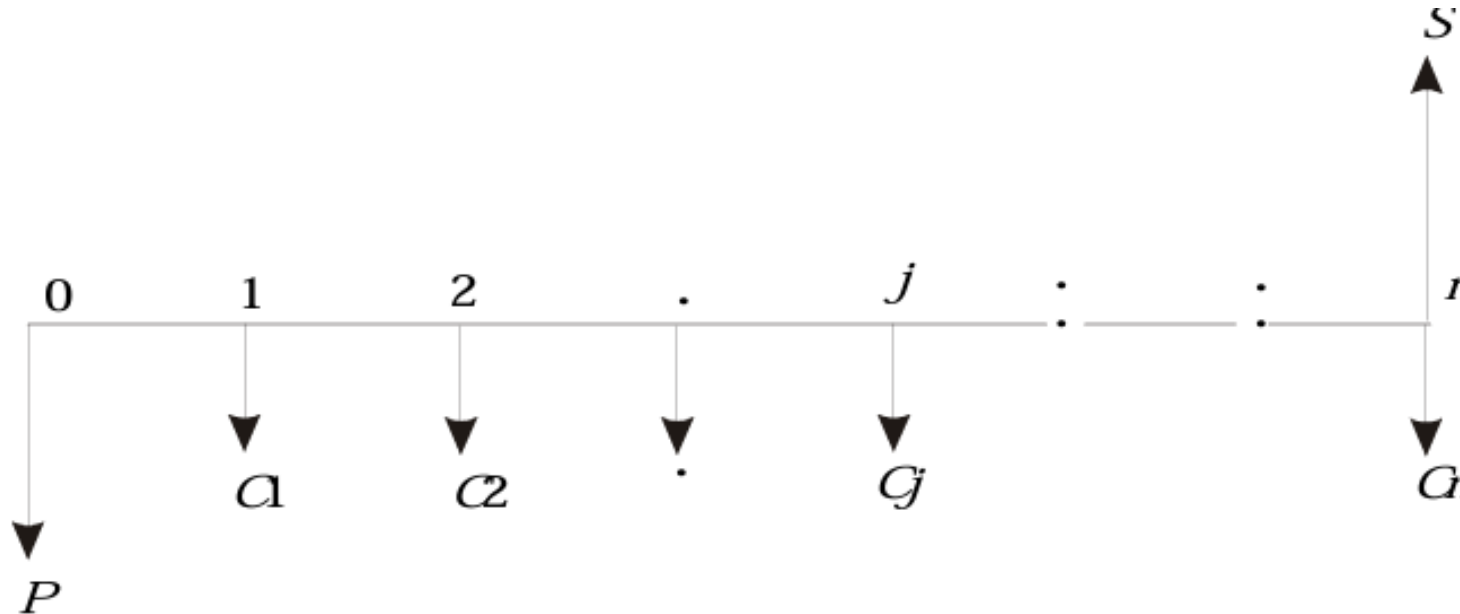


Fig. Cost-dominated cash flow diagram.

Present Worth Method (PW)

P represents an initial investment, C_j the net cost of operation and maintenance at the end of the j th year, and S the salvage value at the end of the n th year. To compute the present worth amount of the above cash flow diagram for a given interest rate i , we have the formula

$$PW(i) = P + C_1[1/(1 + i)^1] + C_2[1/(1 + i)^2] + \dots + C_j[1/(1 + i)^j] \\ + C_n[1/(1 + i)^n] - S[1/(1 + i)^n]$$

In the above formula, the expenditure is assigned a positive sign and the revenue a negative sign. If we have some more alternatives which are to be compared with this alternative, then the corresponding present worth amounts are to be computed and compared. Finally, the alternative with the minimum present worth amount should be selected as the best alternative.

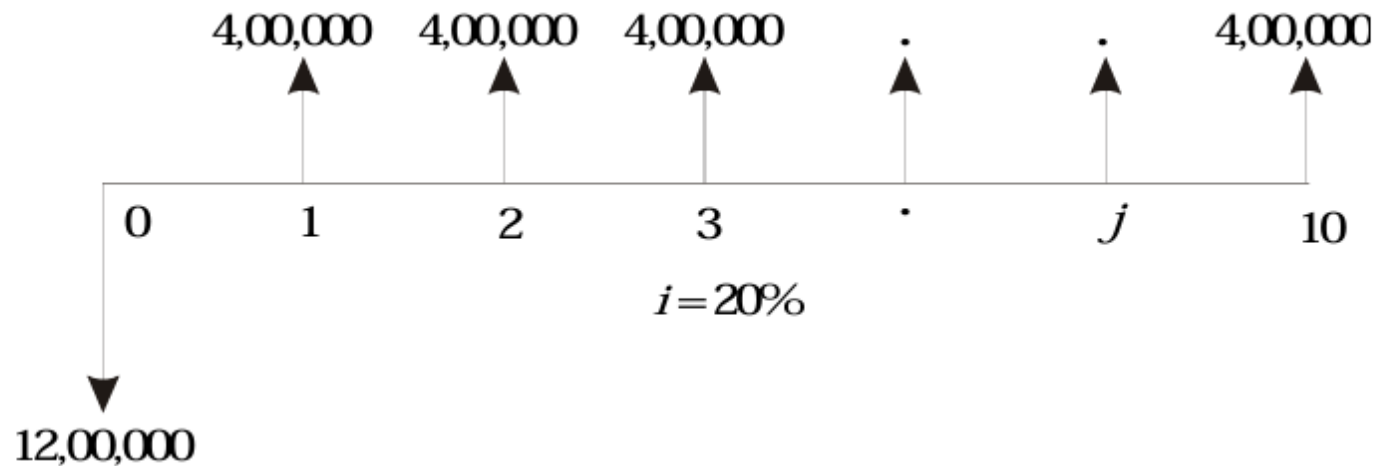
Present Worth Method (PW)

- Alpha Industry is planning to expand its production operation. It has identified three different technologies for meeting the goal. The initial outlay/investment and annual revenues with respect to each of the technologies are summarized in Table. Suggest the best technology which is to be implemented based on the present worth method of comparison assuming 20% interest rate, compounded annually.

	<i>Initial outlay</i> (Rs.)	<i>Annual revenue</i> (Rs.)	<i>Life</i> (years)
Technology 1	12,00,000	4,00,000	10
Technology 2	20,00,000	6,00,000	10
Technology 3	18,00,000	5,00,000	10

Present Worth Method (PW)

TECHNOLOGY I : Initial outlay, $P = \text{Rs. } 12,00,000$, Annual revenue, $A = \text{Rs. } 4,00,000$ Interest rate, $i = 20\%$, compounded annually Life of this technology, $n = 10$ years The cash flow diagram of this technology is as shown in Fig.

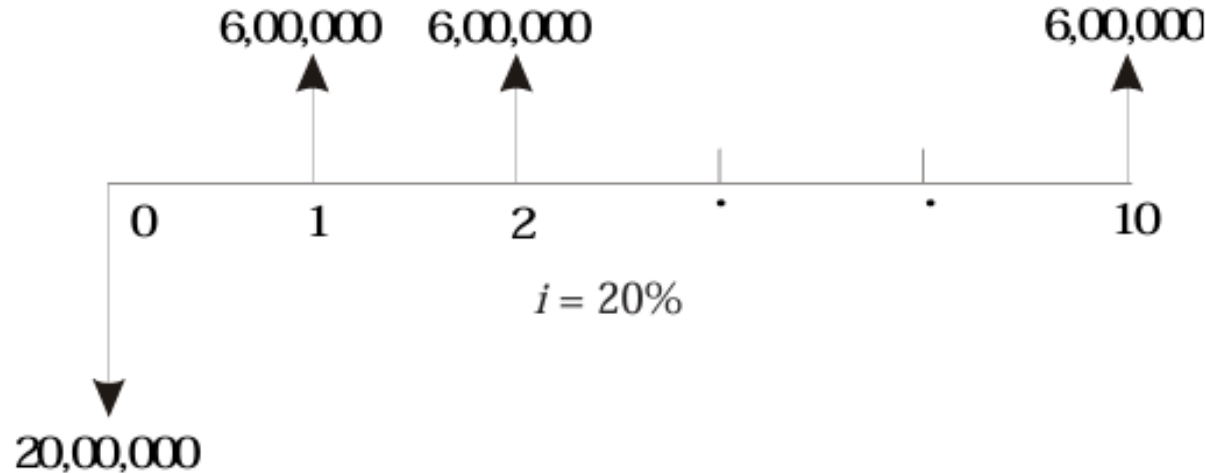


The present worth expression for this technology is $PW(20\%)$
 $= -12,00,000 + 4,00,000 (P/A, 20\%, 10)$
 $= -12,00,000 + 4,00,000 (4.1925)$
 $= -12,00,000 + 16,77,000 = \text{Rs. } 4,77,000$

Present Worth Method (PW)

TECHNOLOGY 2

Initial outlay, $P = \text{Rs. } 20,00,000$ Annual revenue, $A = \text{Rs. } 6,00,000$ Interest rate, $i = 20\%$, compounded annually
Life of this technology, $n = 10$ years The cash flow diagram of this technology is shown in Fig.



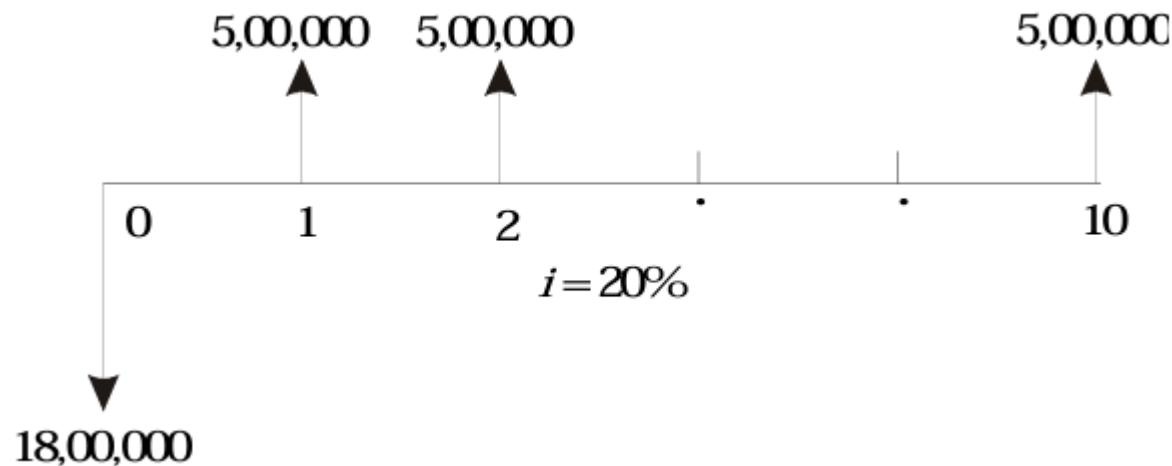
The present worth expression for this technology is $PW(20\%)$
 $= -20,00,000 + 6,00,000 (P/A, 20\%, 10)$
 $= -20,00,000 + 6,00,000 (4.1925)$
 $= -20,00,000 + 25,15,500 = \text{Rs. } 5,15,500 \text{ (Best)}$

Present Worth Method (PW)

TECHNOLOGY 3

Initial outlay, $P = \text{Rs. } 18,00,000$, Annual revenue, $A = \text{Rs. } 5,00,000$ Interest rate, $i = 20\%$, compounded annually
Life of this technology, $n = 10$ years The cash flow diagram of this technology is shown in Fig.

Initial outlay, $P = \text{Rs. } 18,00,000$ Annual revenue, $A = \text{Rs. } 5,00,000$ Interest rate, $i = 20\%$, compounded annually
Life of this technology, $n = 10$ years



The present worth expression for this technology is $PW(20\%)$
 $= -18,00,000 + 5,00,000 (P/A, 20\%, 10)$
 $= -18,00,000 + 5,00,000 (4.1925)$
 $= -18,00,000 + 20,96,250 = \text{Rs. } 2,96,250$

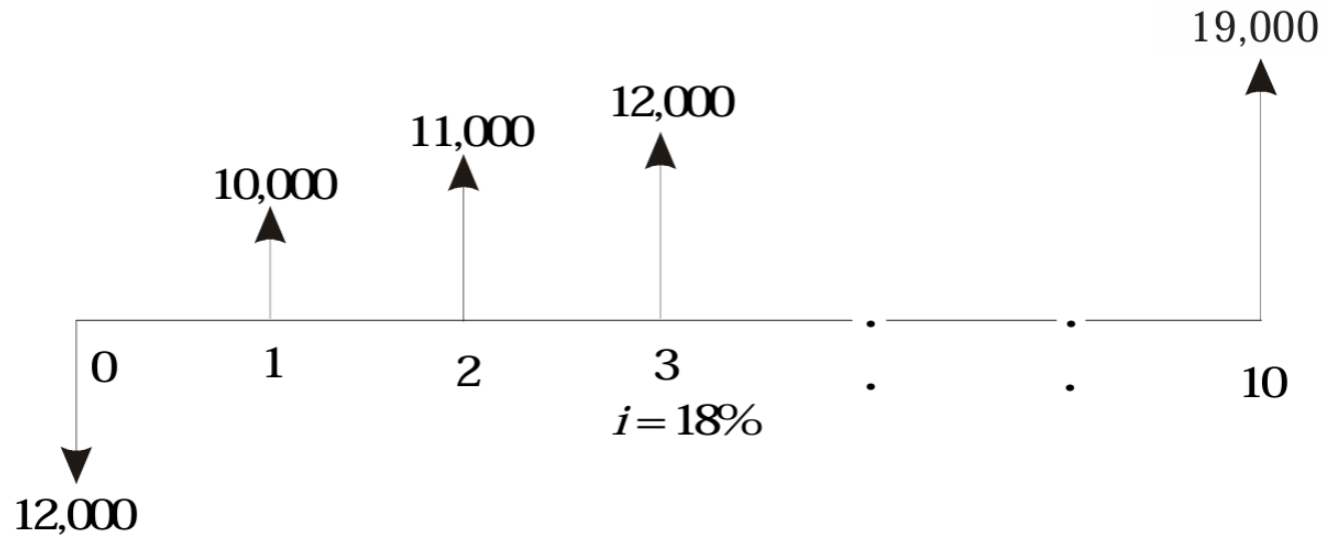
Present Worth Method (PW)

Q) A small business with an initial outlay of Rs. 12,000 yields Rs. 10,000 during the first year of its operation and the yield increases by Rs. 1,000 from its second year of operation up to its 10th year of operation. At the end of the life of the business, the salvage value is zero. Find the present worth of the business by assuming an interest rate of 18%, compounded annually

Initial investment, $P = \text{Rs. } 12,000$ Income during the first year, $A = \text{Rs. } 10,000$ Annual increase in income, $G = \text{Rs. } 1,000$ $n = 10$ years $i = 18\%$, compounded annually

Present Worth Method (PW)

Solution:



Cash flow diagram for the small business.

$$\begin{aligned} \text{The equation for the present worth is } PW(18\%) \\ &= -12,000 + (10,000 + 1,000 (A/G, 18\%, 10)) (P/A, 18\%, 10) \\ &= -12,000 + (10,000 + 1,000 3.1936) 4.4941 \\ &= -12,000 + 59,293.36 = \text{Rs. } 47,293.36 \end{aligned}$$

The present worth of the small business is Rs. 47,293.36.

Present Worth Method (PW)

Q) A granite company is planning to buy a fully automated granite cutting machine. If it is purchased under down payment, the cost of the machine is Rs. 16,00,000. If it is purchased under installment basis, the company has to pay 25% of the cost at the time of purchase and the remaining amount in 10 annual equal installments of Rs. 2,00,000 each. Suggest the best alternative for the company using the present worth basis at $i = 18\%$, compounded annually.

There are two alternatives available for the company:

1. Down payment of Rs. 16,00,000
2. Down payment of Rs. 4,00,000 and 10 annual equal installments of Rs. 2,00,000 each

Present Worth Method (PW)

Present worth calculation of the second alternative. The cash flow diagram of the second alternative is shown

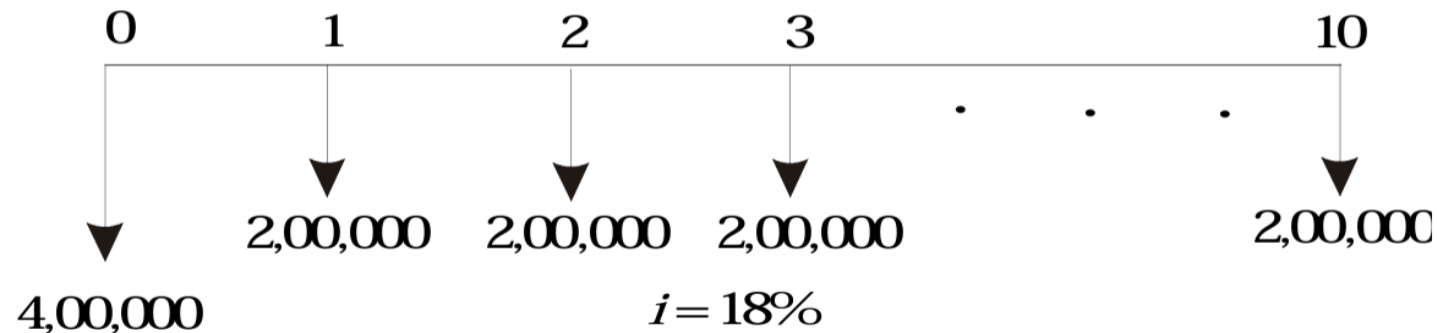


Fig. Cash flow diagram for the second alternative.

The present worth of the above cash flow diagram is computed as

$$\begin{aligned} PW(18\%) &= 4,00,000 + 2,00,000(P/A, 18\%, 10) \\ &= 4,00,000 + 2,00,000 4.4941 \\ &= \text{Rs. } 12,98,820 \end{aligned}$$

- The present worth of this option is Rs. 12,98,820, which is less than the first option of complete down payment of Rs. 16,00,000. Hence, the company should select the second alternative to buy the fully automated granite cutting machine.

Questions

From the above calculations, it is clear that the present worth of technology 2 is the highest among all the technologies. Therefore, technology 2 is suggested for implementation to expand the production.

Q) An engineer has two bids for an elevator to be installed in a new building. The details of the bids for the elevators are as follows:

<i>Bid</i>	<i>Engineer's estimates</i>		
	<i>Initial cost</i> (Rs.)	<i>Service life (years)</i>	<i>Annual operations & maintenance cost (Rs.)</i>
Alpha Elevator Inc.	4,50,000	15	27,000
Beta Elevator Inc.	5,40,000	15	28,500

Determine which bid should be accepted, based on the present worth method of comparison assuming 15% interest rate, compounded annually.

Rs. 6,07,879.80

Questions

Q) Investment proposals A and B have the net cash flows as follows:

<i>Proposal</i>	<i>End of years</i>				
	0	1	2	3	4
A (Rs.)	-10,000	3,000	3,000	7,000	6,000
B (Rs.)	-10,000	6,000	6,000	3,000	3,000

Compare the present worth of A with that of B at $i = 18\%$. Which proposal should be selected?

Rs. 2,767.40

Annual Worth Method (AW)

The **Annual Worth (AW)** method is an economic evaluation technique that converts all project cash flows (costs and benefits) into an equivalent uniform annual amount over the project life at a given Minimum Attractive Rate of Return (MARR).

It is also known as the Equivalent Annual Worth (EAW) or Annual Equivalent (AE) method.

If : $AW > 0$ (accept), $AW = 0$ (Remain indifferent), $AW < 0$ (reject)

Formula 1:

$$AW = R - E - CR$$

Where:

- R = Annual revenue
- E = Annual operating & maintenance cost
- CR = Capital Recovery

$$\text{Where, } CR = P(A/P, i, n) - F(A/F, i, n)$$

OR

Formula 2:

Alternatively, compute the Annual Worth using the formula:

$$AW = -P(A/P, i, n) + (R - E) + F(A/F, i, n)$$

Annual Worth Method (AW)

Capital Recovery (CR):

Generally, there are two types of costs in business operations: operating costs and capital costs.

- Operating costs (labour, raw materials, fuel, etc.) are estimated based on annual production.
- Capital costs (land, buildings, equipment, etc.) occur mainly in the initial period.

For annual equivalent cost analysis, capital costs must be converted into an annual amount over the study period. This annual equivalent capital cost is called **Capital Recovery (CR)**.

Capital recovery covers:

Depreciation of capital assets

Interest on invested money

A capital asset includes the initial investment (I) and the salvage value (S) at the end of its useful life.

Annual Worth Method (AW)

Capital Recovery Formula

$$CR(i) = I(A/P, i\%, N) - S(A/F, i\%, N)$$

Alternative Forms

$$CR(i) = (I - S)(A/P, i\%, N) + S(i)$$

or

$$CR(i) = (I - S)(A/F, i\%, N) + I(i)$$

Annual Worth Method (AW)

Q) A machine requires an initial investment of Rs. 1,00,000 and has a useful life of 10 years with a salvage value of Rs. 25,000 at the end of its life. The machine generates an annual revenue of Rs. 20,000 and incurs annual operating and maintenance expenses of Rs. 5,000. Determine the investment decision using the Annual Worth method.

Also calculate the Capital Recovery (CR) of the project

Capital Recovery Calculation

$$CR(10\%) = I(A/P, 10\%, 10) - S(A/F, 10\%, 10)$$

Substituting values:

$$CR = 1,00,000(0.1627) - 25,000(0.06274)$$

$$CR = 16,270 - 1,568$$

$$CR = \text{Rs. } 14,702$$

Annual Worth Method (AW)

For Annual Worth (AW):

$$AW(10\%) = R - E - CR$$

$$AW(10\%) = 20,000 - 5,000 - 14,702$$

$$AW(10\%) = Rs. 298$$

Alternatively,

$$AW(10\%) = -1,00,000(A/P, 10\%, 10) + (20,000 - 5,000) + 25,000(A/F, 10\%, 10)$$

$$AW(10\%) = -1,00,000(0.1627) + 15,000 + 25,000(0.06274)$$

$$AW(10\%) = -16,270 + 15,000 + 1,568$$

$$AW(10\%) = Rs. 298$$

Since $AW(10\%) > 0$, the investment is acceptable.

Future Worth Method (FW)

The **Future Worth method** is an economic evaluation technique in which all cash inflows and outflows are converted to their equivalent value at the end of the project life using the Minimum Attractive Rate of Return (MARR). If : $FW > 0$ (accept), $FW = 0$ (Remain indifferent), $FW < 0$ (reject)

1. Single Present Amount

$$F = P(F/P, i, n) = P(1 + i)^n$$

2. Uniform Annual Cash Flow

$$F = A(F/A, i, n)$$

3. Complete FW Equation

$$FW = -P(F/P, i, n) + A(F/A, i, n) + S$$

$$FW = -P(1 + i)^n + A \left(\frac{(1 + i)^n - 1}{i} \right) + S$$

Future Worth Method (FW)

Parameter	Value (Rs)
Initial Investment (I)	50,000
Annual Revenue (R)	18,000
Annual Cost (C)	4,000
Life of Machine (n)	5 years
Salvage Value (S)	8,000
MARR	10%

$$FW = -P(1 + i)^n + A \left(\frac{(1 + i)^n - 1}{i} \right) + S$$

$$FW = (18,000 - 4,000) \frac{(1.1)^5 - 1}{0.1} - 50,000(1.1)^5 + 8,000$$

12,945 > 0 (Accept)

Payback Period

Payback Period is the time required for an investment to recover its initial cost from the net cash inflows generated by the project. In other words, it tells how long it takes for a project to “pay back” the money invested.

Types of Payback period:

a) Simple Payback Period

- Considers nominal cash inflows only.
- Ignores the time value of money.

$$\text{PBP (years)} = \frac{\text{Initial Investment (I)}}{\text{Annual Net Cash Inflow (R - C)}}$$

Formula (Non-uniform Cash Flow)

$$\text{Payback Period} = \text{Years before recovery} + \frac{\text{Unrecovered cost}}{\text{Cash flow in recovery year}}$$

Payback Period

(b) Discounted Payback Period (DPP)

- Considers time value of money
- Uses discounted cash flows
- More accurate than simple payback

$$\text{Solve } \sum_{t=1}^n \frac{CF_t}{(1+i)^t} = I$$

Formula (Non-uniform Cash Flow)

$$\text{Payback Period} = \text{Years before recovery} + \frac{\text{Unrecovered cost}}{\text{Cash flow in recovery year}}$$

Questions

End of period	Net cash flow (Rs.)	Cumulative cash flow (Rs.)
0	-25,000	-25,000
1	+8,000	-17,000
2	+8,000	-9,000
3	+8,000	-1,000
4	+8,000	+7,000
5	+13,000	+20,000

Here, cumulative cash flow turns to positive in period 4. Therefore, payback period lies between period 3 and 4. By interpolating, we get payback period

$$\text{Simple Payback Period} : 3 + \frac{1,000}{8,000} = 3.125 \text{ periods}$$

Questions

End of period	Net cash flow (Rs.)	Discounted cash flow into present (Rs.) @20%	Cumulative cash flow (Rs.)
0	-25,000	-25,000	-25,000
1	+8,000	6,667	-18,333
2	+8,000	5,556	-12,777
3	+8,000	4,630	-8,147
4	+8,000	3,858	-4,289
5	+13,000	5,224	+935

Here, the cumulative cash flow turns to positive in period 5. Therefore, payback period lies between period 4 and 5. By interpolating, we get the required payback period:

$$4 + \frac{4,289}{5,224} = 4.82 \text{ periods}$$

Public Sector Economic Analysis/ Benefit Cost Analysis

Benefit-Cost analysis is an economic evaluation technique used to determine the economic feasibility of a project by comparing the benefits it produces with the costs it incurs. It is widely used in engineering, public projects, and capital budgeting.

Purpose:

To decide whether a project is worth investing in

To compare multiple alternatives and select the most economical one

To optimize resource allocation in public or private investments

Types of Benefit–Cost Analysis

- a. Conventional BCR
- b. Modified BCR

Public Sector Economic Analysis/ Benefit Cost Analysis

I) Conventional Benefit–Cost Ratio Method

Present Worth (PW)

$$\text{BCR}_{\text{Conventional}} = \frac{P_B}{P_I + P_{O\&M} - P_S}$$

Future Worth (FW)

$$\text{FW-BCR} = \frac{F_B}{F_I + F_{O\&M} - F_S}$$

Annual Worth (AW)

$$\text{AW-BCR} = \frac{A_B}{A_I + A_{O\&M} - A_S}$$

Benefit–Cost Analysis

2) Modified Benefit–Cost Ratio method

$$\text{BCR}_{\text{PW, Modified}} = \frac{P_B - P_{O\&M}}{P_I - P_S}$$

$$\text{BCR}_{\text{FW, Modified}} = \frac{F_B - F_{O\&M}}{F_I - F_S}$$

$$\text{BCR}_{\text{AW, Modified}} = \frac{A_B - A_{O\&M}}{A_I - A_S}$$

Decision Rule:

- $\text{BCR} > 1$ (Accept)
- $\text{BCR} < 1$ (Reject)
- $\text{BCR} = 1$ (Remain Indifferent)

Internal Rate of Return (IRR)

The **Internal Rate of Return (IRR)** is the interest rate at which the net present worth (NPW) of all cash flows associated with a project becomes zero.

It represents the actual earning power of an investment.

$$\text{NPW}(i) = 0$$

- IRR is the break-even interest rate of a project.
- It is compared with the Minimum Attractive Rate of Return (MARR) to decide project acceptance.
- If $\text{IRR} \geq \text{MARR}$, the project is economically acceptable.
- If $\text{IRR} < \text{MARR}$, the project is rejected.

Internal Rate of Return (IRR)

For a project with an initial investment P and net cash flows CF_t at the end of year t :

$$-P + \sum_{t=1}^n \frac{CF_t}{(1+i)^t} = 0$$

Where:

- P = Initial investment
- CF_t = Net cash flow in year t
- n = Project life (years)
- i = Internal Rate of Return (IRR)

Internal Rate of Return (IRR)

Interpolation Formula for IRR

When NPW is calculated at two interest rates i_1 and i_2 :

$$\text{IRR} = i_1 + \left(\frac{\text{NPW}_{i_1}}{\text{NPW}_{i_1} - \text{NPW}_{i_2}} \right) (i_2 - i_1)$$

Where:

- NPW_{i_1} = Positive NPW at lower rate
- NPW_{i_2} = Negative NPW at higher rate

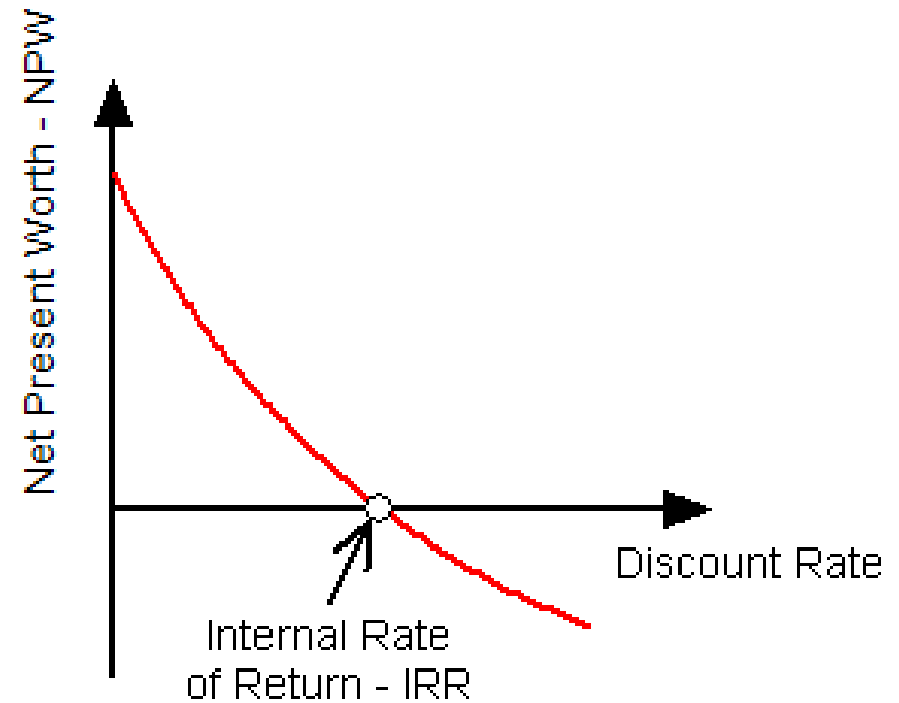
Internal Rate of Return (IRR)

Advantages of IRR

- Considers time value of money
- Easy to interpret as a percentage return
- Widely used in engineering project evaluation

Limitations of IRR

- Requires trial-and-error or numerical methods
- May give multiple IRRs
- Assumes reinvestment at IRR (often unrealistic)
- Not suitable alone for mutually exclusive projects



Internal Rate of Return (IRR)

Calculate IRR from the following cash flow and draw investment balance diagram.

Year	0	1	2	3	4	5
Cash Flow	−800	250	300	400	−150	600

Solution:

$$PW = -800 + 250(1 + i)^{-1} + 300(1 + i)^{-2} + 400(1 + i)^{-3} - 150(1 + i)^{-4} + 600(1 + i)^{-5}$$

For $PW = 0$, solve for i .

Trial Values

For $i = 20\%$:

$$PW = 17$$

For $i = 21\%$:

$$PW = -1.3$$

Interpolation Formula

$$IRR = i_1 + \left(\frac{PW_1}{PW_1 - PW_2} \right) (i_2 - i_1)$$

$$IRR = 20\% + \left(\frac{17}{17 - (-1.3)} \right) (1\%)$$

$$IRR = 20\% + \frac{17}{18.3}$$

$$IRR = 20.93\%$$

Unrecovered Investment Calculation (IRR = 20.93%)

EOY	Cash Flow	Unrecovered Investment (Beginning of Year)	End of Year @ 20.93%	Unrecovered Investment
0	−800	−	−800	−800
1	250	−800	−967	−717
2	300	−717	−867	−567
3	400	−567	−686	−286
4	−150	−286	−346	−496
5	600	−496	−600	0

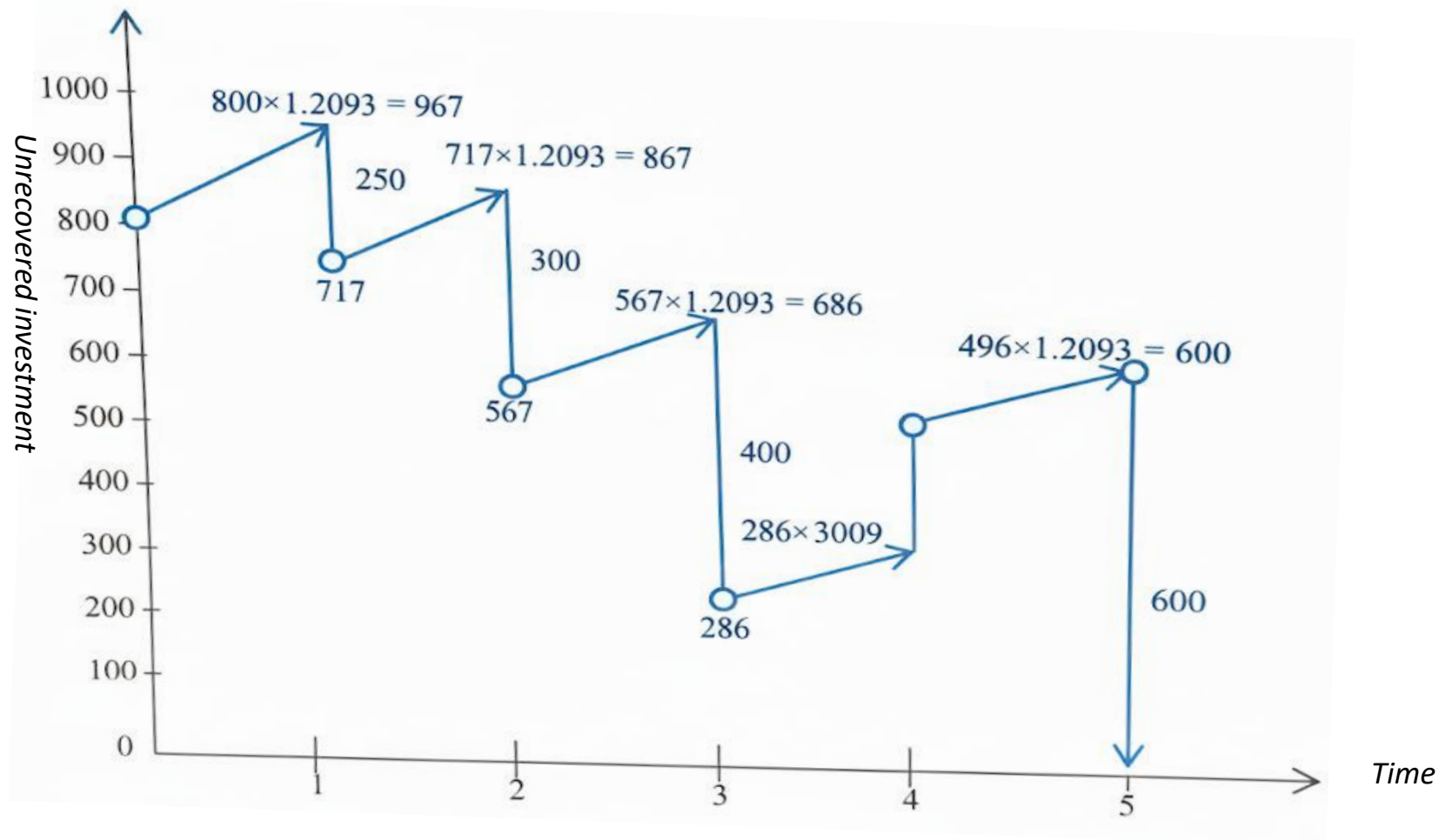


Fig: Unrecovered/ Investment balance diagram

The unrecovered balance diagram graphically represents the year-wise recovery of investment at IRR and verifies that the unrecovered balance becomes zero at the end of the project life.

External Rate of Return (ERR)

The **External Rate of Return (ERR)** is the rate of return on a project assuming that all intermediate positive cash flows are reinvested at a specified external rate, usually the Minimum Attractive Rate of Return (MARR), instead of at the IRR.

- All negative cash flows are compounded to present worth at the external rate
- All positive cash flows are compounded to future worth at
-
- ERR is the rate that equates these two values over the project life

ERR Formula

$$PW_{\text{costs}}(1 + \text{ERR})^n = FW_{\text{benefits}}$$

Questions

A civil engineering project has the following uneven cash flow. The external reinvestment rate (MARR) is 10% per year. Evaluate using ERR

Year	0	1	2	3	4	5
Cash Flow (Rs.)	-50,000	+12,000	+15,000	-8,000	+20,000	+25,000

Questions

Step 1: Separate costs and benefits

2: Discount costs to Year 0 at external rate = 10%

Costs (-):

- Year 0: -50,000
- Year 3: -8,000

$$PW_{\text{costs}} = 50,000 + \frac{8,000}{(1.10)^3}$$

Benefits (+):

- Year 1: 12,000
- Year 2: 15,000
- Year 4: 20,000
- Year 5: 25,000

$$PW_{\text{costs}} = 50,000 + 6,010$$

$$PW_{\text{costs}} = 56,010$$

Questions

3: Future Worth (FW) of all benefits . Compound benefits to Year 5 at 10%

$$FW_{\text{benefits}} = 12,000(1.10)^4 + 15,000(1.10)^3 + 20,000(1.10)^1 + 25,000$$

$$FW_{\text{benefits}} = 17,570 + 19,965 + 22,000 + 25,000$$

$$FW_{\text{benefits}} = 84,535$$

$$PW_{\text{costs}}(1 + \text{ERR})^n = FW_{\text{benefits}}$$

$$56,010(1 + \text{ERR})^5 = 84,535$$

$$\text{ERR} = 8.5 \%$$

Since ERR is less than MARR, the project should be rejected.

Repeatability assumption

The **repeatability method** is an economic comparison technique used for evaluating mutually exclusive alternatives with unequal useful lives, based on the assumption that each alternative can be repeated indefinitely with identical costs, benefits, and service life.

- It assumes that when an alternative reaches the end of its useful life, it is replaced by an identical alternative under the same economic and operating conditions.
- This assumption allows all alternatives to be compared over a common study period, equal to the least common multiple (LCM) of their individual lives.

Repeatability assumption / Steps

1. Find common study period

$$N = \text{LCM}(n_1, n_2, \dots)$$

2. Repeat projects to cover N years

Number of repetitions:

$$\text{Repetitions} = \frac{N}{n}$$

3. Compute net annual cash flow

$$A = A_r - A_c$$

4. Present worth of initial & repeated investments

$$PWC = \sum C_0(1+i)^{-t}$$

5. Present worth of salvage values

$$PWS = \sum S(1+i)^{-t}$$

6. Present worth of annual cash flows

$$PW_{\text{annual}} = A \left[\frac{(1+i)^N - 1}{i(1+i)^N} \right]$$

7. Total present worth of project

$$PW = PWC + PWS + PW_{\text{annual}}$$

Decision rule

Select project with **maximum PW** at given i .

Repeatability assumption

Q) By using PW or FW and Repeatability method select the best project.

Item	Project A	Project B
Initial Investment	Rs. 4,00,000	Rs. 6,00,000
Annual Revenue	Rs. 30,000	Rs. 35,000
Annual O&M Cost	Rs. 3,000	Rs. 4,000
Useful Life	6 years	8 years
Salvage Value	Rs. 4,000	Rs. 7,000
MARR	12%	12%

Co-Terminated assumptions

The **co-terminated assumption** is a method used to compare mutually exclusive alternatives (MEAs) that have unequal useful lives. Unlike the repeatability assumption which assumes a project will be repeated over a common multiple of lives the co-terminated assumption uses a fixed, finite, and identical study period for all alternatives.

The co-terminated assumption compares alternatives with unequal lives over a common study period, by terminating longer-life alternatives at the end of the study period and accounting for their imputed market (salvage) value.

Co-Terminated assumptions

1. Select a common study period (N)
(Fixed analysis horizon; LCM is not used)
2. Compute PW of alternatives whose life = N

$$PW = -P + A(P/A, i, N) + S(P/F, i, N)$$

3. For alternatives with life > N, terminate at year N
4. Calculate Capital Recovery (CR)

$$CR = P(A/P, i, n) - S(A/F, i, n)$$

➤ *Capital Recovery is the uniform annual equivalent cost of owning an asset over its entire economic life, considering initial cost and salvage value.*

Co-Terminated assumptions

5. Find Imputed Market Value (IMV) at year N

$$IMV = CR(P/A, i, \text{remaining life})$$

6. Compute PW including IMV as salvage

$$PW = -P + A(P/A, i, N) + IMV(P/F, i, N)$$

7. Compare PW values and select maximum PW

➤ *Imputed Market Value is the economic value of a project at the end of the study period, equal to the present worth of the remaining capital recovery payments.*

Co-Terminated assumptions

Q) Recommend the best project from the following information using: Co-termination assumption. If (MARR) is 12% per year.

Project	A	B
Initial Investment (I)	4,00,000	7,00,000
Annual Revenue (AR)	1,75,000	2,50,000
Annual Cost (AC)	25,000	25,000
Salvage Value (S)	40,000	70,000
Project Life (N)	6 years	8 years

PW equation used:

$$PW_A = -P + A(P/A, i, n) + S(P/F, i, n)$$

Substitution:

$$PW_A = -4,00,000 + 1,50,000(P/A, 12\%, 6) + 40,000(P/F, 12\%, 6)$$

Using factor values:

- $(P/A, 12\%, 6) = 4.1114$
- $(P/F, 12\%, 6) = 0.5066$

$$PWA = 2,36,974$$

Now,

Project B's actual life = 8 years

Study period chosen = 6 years

So Project B must be terminated at year 6

We must find what Project B is worth at the end of year 6

➤ Compute Capital Recovery (CR) of Project B

$$CR = P(A/P, i, n) - S(A/F, i, n)$$

Initial cost = 7,00,000

Salvage at year 8 = 70,000

Life = 8 years

Interest = 12%

$$CR = 7,00,000(0.2013) - 70,000(0.0813)$$

$$CR = 1,35,219 \approx 1,35,000$$

Imputed Market Value (Salvage at Year 6)

Since Project B is stopped 2 years before its full life, its value at year 6 is the present worth (at year 6) of the remaining capital recovery payments.

$$IMV = CR(P/A, i, 2)$$

$$IMV = 1,35,000 \times 1.6901$$

$$IMV = 2,28,534$$

PW of Project B over 6 Years

- Initial cost = 7,00,000
- Annual benefit = 2,15,000
- Salvage (IMV) at year 6 = 2,28,534

$$PW_B = -P + A(P/A, i, n) + IMV(P/F, i, n)$$

Substitution:

$$PW_B = -7,00,000 + 2,15,000(4.1114) + 2,28,534(0.5066)$$

Result:

$$PW_B = 2,99,726$$

$PW_B > PWA$

Project B is economically preferred

Weighted average cost of capital

WACC is the average rate of return a company is expected to pay to all its security holders (equity holders, debt holders, preferred shareholders), weighted by their proportion in the company's capital structure.

It represents the minimum return a company must earn on its existing assets to satisfy its investors.

Cost of Equity (R_e): Represents the return shareholders expect, typically calculated using the CAPM(Capital Asset Pricing Model)

formula: $R_e = R_f + \beta \times (R_m - R_f)$, where R_f is the risk-free rate, β is beta, and R_m is the expected market return.

Cost of Debt (R_d): The interest rate paid on borrowings, adjusted for the tax shield since interest payments are tax-deductible. The after-tax cost is $R_d \times (1 - T_c)$.

Weights (E/V and D/V): These proportions reflect the company's capital structure based on market values.

Strategic Importance

Weighted average cost of capital

Formula for WACC

$$WACC = \frac{E}{V} \cdot R_e + \frac{D}{V} \cdot R_d \cdot (1 - T)$$

WACC < MARR / IRR (Accept)

(E)	Market value of equity
(D)	Market value of debt
(V = E + D)	Total value of capital (equity + debt)
(Re)	Cost of equity (expected return by equity holders)
(Rd)	Cost of debt (interest rate on debt)
(T)	Corporate tax rate

Weighted average cost of capital

WACC is crucial for:

- Evaluating Investments: Projects with a projected Internal Rate of Return (IRR) higher than WACC are typically seen as value-adding.
- Valuing Businesses: It is used as the discount rate in Discounted Cash Flow (DCF) analysis.
- Assessing Risk: A higher WACC can indicate a higher-risk business

Q) Consider a company with \$800M in equity and \$200M in debt (total \$1B), a 12% cost of equity, 5% cost of debt, and 25% tax rate.

- Equity weight: $\$800\text{M} / \$1,000\text{M} = 0.8$
- Debt weight: $\$200\text{M} / \$1,000\text{M} = 0.2$
- After-tax cost of debt: $5\% \times (1 - 0.25) = 3.75\%$
- WACC: $(0.8 \times 12\%) + (0.2 \times 3.75\%) = 9.6\% + 0.75\% = 10.35\%$

Weighted average cost of capital

Q) A private firm is awarded a road construction project. The investment is financed by:

- Equity: Rs. 80 lakh at 20%
- Debt: Rs. 1.2 crore at 10%

Income tax rate = 25%.

Compute the WACC.

State whether the project is acceptable if the expected return is 14%.

Financial and economic analysis

Financial Analysis	Economic Analysis
Evaluation of a project from the investor/firm's point of view	Evaluation of a project from the society/nation's point of view
To check profitability and financial feasibility	To check overall economic desirability and social welfare
Private (owner, company, investor)	Public (society, government, economy)
Only actual cash costs (capital, O&M, taxes, interest)	All real costs, including social and opportunity costs
Only direct monetary benefits	Direct + indirect benefits (employment, environment, regional growth)
Market prices (with taxes, subsidies, duties)	Shadow prices / economic prices (taxes & subsidies excluded)
Included (real cash flows)	Excluded (transfer payments, not real resource use)
Ignored	Considered (pollution, health impact, social effects)
Market interest rate / MARR / cost of capital	Social discount rate
NPV, IRR, BCR, Payback Period	Economic NPV, Economic IRR, Economic BCR
Firms, private investors, banks	Government, planners, policy makers

Multiple investment project alternatives (Dependent, independent and contingent)

Investment project alternatives are typically classified based on their relationship with other potential investments. These classifications determine the decision-making rules used in capital budgeting.

I. Independent Projects

Independent projects are those whose cash flows are not affected by the acceptance or rejection of other projects.

- Key Characteristic: The decision to invest in one project has no bearing on the decision for another; they do not compete for the same physical or strategic space.
- Selection Rule: A firm should ideally accept all independent projects that meet a minimum performance metric, such as a positive Net Present Value (NPV) or FW or AW provided there is enough capital.

Multiple investment project alternatives (Dependent, independent and contingent)

2. Dependent projects

Dependent projects are those where the acceptance of one proposal is directly linked to the decision regarding another. These are generally divided into two sub-categories:

- **Mutually Exclusive Projects:** These projects serve the same purpose and compete directly.

Accepting one automatically precludes the acceptance of the others.

- **Selection Rule:** Only the best alternative is chosen (typically the one with the highest NPV).

3. Contingent projects

Contingent projects are a specific type of dependent project where the acceptance of one project is required for another to proceed, or one project serves as a prerequisite.

- **Key Characteristic:** These cannot be undertaken singularly; they must be completed together or in a specific sequence.

- **Selection Rule:** They are often analyzed as a single package or "combined project" because the primary project cannot function without the contingent one.

Example: Building a factory in a remote area is the primary project, while the construction of roads, employee housing, and schools for that site are contingent projects.

Assignment

1. A project involves an initial outlay of Rs. 30,00,000 and with the following transactions for the next five years. The salvage value at the end of the life of the project after five years is Rs. 2,00,000. Draw a cash flow diagram of the project and find its present worth by assuming $i = 15\%$, compounded annually.

<i>End of year</i>	<i>Maintenance and operating expense (Rs.)</i>	<i>Revenue (Rs.)</i>
1	2,00,000	9,00,000
2	2,50,000	10,00,000
3	3,00,000	12,00,000
4	3,00,000	13,00,000
5	4,00,000	12,00,000

2) Check the feasibility of the following investment using Present Worth (PW), Annual Worth (AW), and Future Worth (FW) methods. Also determine the capital recovery amount of the project.

- First investment = Rs. 20,00,000
- Salvage value = Rs. 50,000
- Project period = 10 years
- Annual revenue = Rs. 50,000
- Annual cost = Rs. 15,000
- MARR = 10% per year

3) Evaluate the following project on the basis of Present Worth (PW) and Future Worth (FW) methods when MARR = 10%.

First investment = Rs. 15,00,000

Salvage value = Rs. 35,000

Project life = 15 years

Annual expenses = Rs. 40,000

Annual revenue = Rs. 25,000

Overhead cost at the end of 5th year = Rs. 50,000

Overhead cost at the end of 10th year = Rs. 50,000

4) Evaluate the following project using the Internal Rate of Return (IRR) method when $MARR = 12\%$. State whether the project is acceptable or not.

End of Year	Cash Flow (Rs.)
0	-2,50,000
1	25,000
2	50,000
3	75,000
4	1,00,000
5	1,25,000

5) Evaluate the following cash flow using the IRR method and identify the investment decision. Also draw the investment balance diagram.

First investment = Rs. 5,00,000

Annual revenue = Rs. 1,20,000

Annual cost = Rs. 30,000

Useful life = 10 years

6) Find the External Rate of Return (ERR) of the following project and determine whether the project is acceptable when $MARR = 15\%$.

- Initial investment = Rs. 30,000
- Salvage value = Rs. 2,000
- Useful life = 5 years
- Annual revenue = Rs. 12,000

7) The cash flows of Project B are given below. Calculate the discounted payback period using $i = 15\%$ per year and comment on the feasibility of the project.

End of Year	Cash Flow (Rs.)
0	-4,00,000
1	1,10,000
2	1,10,000
3	1,10,000
4	1,10,000
5	1,10,000

8) The cash flows of Project C are given below. Determine both types of payback period at $i = 15\%$ and comment on the acceptability of the project.

End of Year	Cash Flow (Rs.)
0	-7,00,000
1	1,70,000
2	1,70,000
3	1,70,000
4	1,70,000
5	3,10,000

9) Compare Project A and Project B using the repeatability assumption and Co-terminated method , when MARR = 10%.

Particulars	Project A	Project B
Initial investment (Rs.)	1,00,000	1,50,000
Annual revenue (Rs.)	15,000	20,000
Annual cost (Rs.)	3,000	4,000
Useful life (years)	6	8
Salvage value (Rs.)	5,000	10,000

10) Two mutually exclusive projects A and B are proposed. The following data are given:

Project	Initial Investment	Annual O&M	Annual Benefit	Salvage	Life (years)
A	2,00,000	15,000	50,000	20,000	5
B	1,50,000	10,000	40,000	10,000	5

- Calculate Conventional BCR for both projects using PW, FW, AW method.
- Calculate Modified BCR for both projects using PW, FW, AW method.
- Decide which project should be selected.

11) Compare the following two mutually exclusive projects using: (i) Co-terminated assumption, and (ii) Repeatability assumption.

Take MARR = 8%.

	Project A	Project B
Initial cost (Rs.)	1,50,000	2,00,000
Annual revenue (Rs.)	90,000	1,00,000
Operating cost (Rs.)	20,000	20,000
Life (years)	4	6
Salvage value (Rs.)	80,000	1,20,000

12) A firm undertakes a small hydropower project. The capital structure is:

- Equity = 40% at 22%
- Debt = 60% at 9%
- Tax rate = 30%.
- Calculate WACC.

If the project IRR is 13%, should the project be accepted?

13) A contractor plans to construct a commercial building. The project is financed as follows:

- 60% from bank loan at 12% annual interest
- 40% from owner's capital expecting 18% return

If the corporate tax rate is 30%,
calculate the Weighted Average Cost of Capital (WACC) for the project.

Thank you for your attention!